Supplementary Materials
Hyperelastic Regularization for Reducing Folding in Transformer-Based Brain MRI Registration

This document accompanies the manuscript *Hyperelastic Regularization for Reducing Folding in Transformer-Based Brain MRI Registration*. Contents include training hyperparameters and evaluation notes, IXI baseline checkpoint summary (Table S1), auxiliary IXI deformation/surface metrics omitted from the main comparison table (Table S2), paired inferential exports on IXI (Tables S3–S4 under *Inferential Statistics*), grouped-VOI mapping (Table S5), Figure S1 (descriptive IXI bar-chart overview), Figure S2 (qualitative registration comparison across three orthogonal views), OASIS training notes, OASIS-retrained contrasts versus selected baselines (Table S6), and bootstrap 95% confidence intervals on IXI means (Table S7).

1 Supplementary Experimental Details

1.1 Model and Environment Configuration

The following settings are sufficient to reproduce HypEReg-TransMorph training on the IXI atlas-to-subject protocol with the public TransMorph-style backbone.

Software. Python 3.12.x; PyTorch (CUDA build matched to the GPU). Classical baselines and distance transforms use antspyx and SimpleITK as in the main evaluation. The reported experiments used PyTorch 2.12 (CUDA 12.8) on one NVIDIA RTX PRO 6000 (97 GiB VRAM); other CUDA-capable GPUs are suitable (reduce batch size if memory-limited).

Data protocol. Preprocessed IXI T1 volumes in template space, spatial size $160 \times 192 \times 224$, with the standard 403/58/115 train/validation/test split and a single fixed atlas. Training batch size 2; validation batch size 1. Training augmentations: independent random flips along each spatial axis with probability 0.5; cast image pairs to float32. Validation: FreeSurfer-compatible label relabeling to the 46-label index set used in this study, then float32 images and int16 labels.

Optimization. Adam with AMSGrad, learning rate 10^{-4}, weight decay 0, and PyTorch defaults $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$. Train for 500 epochs with bidirectional updates ($x \to y$ and $y \to x$): one forward–backward–optimizer step per direction per mini-batch. Learning-rate decay each iteration:

$$\mathrm{lr}(e) = \mathrm{lr}_0 \left(1 - \frac{e}{500}\right)^{0.9}.$$

Retain the checkpoint with the best validation grouped Dice. As in the upstream TransMorph reference script, a single global random seed is not pinned; the released weights define the reference run (see main-text limitations on multi-seed retraining).

Loss. Total loss per direction (unit weights on all three terms):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{grad}} + \mathcal{L}_{\text{HypEReg}}.$$

Similarity $\mathcal{L}_{\text{sim}} = -\overline{\text{LCC}}^{2}$ with a 9^3 local window and denominator stabilizer 10^{-5}. Smoothness $\mathcal{L}_{\text{grad}}$: mean squared first-order finite differences of the displacement along x, y, and z, averaged over the three axes. HypEReg weights $(\beta, \gamma, \epsilon) = (0.02, 20, 10^{-3})$; BMR-style length/surface terms are disabled ($\alpha_{\text{length}} = 0$). Jacobian $J_\phi = I + \nabla u$ from forward finite differences on interior voxels; volume term $(\det J_\phi - 1)^2 / \max(\det J_\phi, \epsilon)$; folding term $[\max(0, \epsilon - \det J_\phi)]^2$ (equivalently $\text{ReLU}(\epsilon - \det J_\phi)^2$).

TransMorph backbone. Patch size 4; input channels 2; spatial size $(160, 192, 224)$; embed dimension 96; encoder depths $(2, 2, 4, 2)$; attention heads $(4, 4, 8, 8)$; window sizes $(5, 6, 7, 7)$; MLP ratio 4; patch-merging reduction factor 4; registration-head channels 16; skip connections from Transformer stages and from the encoder enabled; `qkv_bias` = `False`; dropout 0; stochastic depth rate 0.3; relative position bias enabled; patch normalization on; absolute/relative sinusoidal PE off; gradient checkpointing off; decoder feature indices $(0, 1, 2, 3)$. The released HypEReg-TransMorph and TransMorph checkpoints instantiate $\sim$46.77 M parameters.

Forward runtime profiling (main text). Profile in no-gradient mode, input shape $1 \times 2 \times 160 \times 192 \times 224$, five warm-up iterations and twenty timed repeats; report mean wall time and peak device memory. On the hardware above, HypEReg-TransMorph and TransMorph yield 0.0822 s and 5.685 GB; TransMorphBayes yields 0.0852 s and 6.042 GB.

Table S1: Baseline checkpoints used for IXI evaluation (identifiers only; full weight files ship with the software archive and data release).

Model	**Checkpoint / source note**
HypEReg-TransMorph	HypEReg-TransMorph; validation-selected IXI run (HypEReg weights β=0.02, γ=20).
TransMorph	Public TransMorph IXI checkpoint from the paper release (validation DSC $\approx$ 0.744).
TransMorphBayes	Bayesian TransMorph checkpoint from the same release (validation DSC $\approx$ 0.743).
MIDIR	Released MIDIR validation checkpoint (DSC $\approx$ 0.733).
CycleMorph	Released CycleMorph validation checkpoint (DSC $\approx$ 0.729).
VoxelMorph-1	Released single-resolution VoxelMorph checkpoint (DSC $\approx$ 0.720).
CoTr	Released CoTr checkpoint (DSC $\approx$ 0.730).
nnFormer	Released nnFormer checkpoint (DSC $\approx$ 0.739).
PVT	Released PVT checkpoint (DSC $\approx$ 0.720).
SyN (ANTs)	Classical SyN outputs under the shared IXI protocol (no network weights).

2 Figure S1: Descriptive IXI Overview

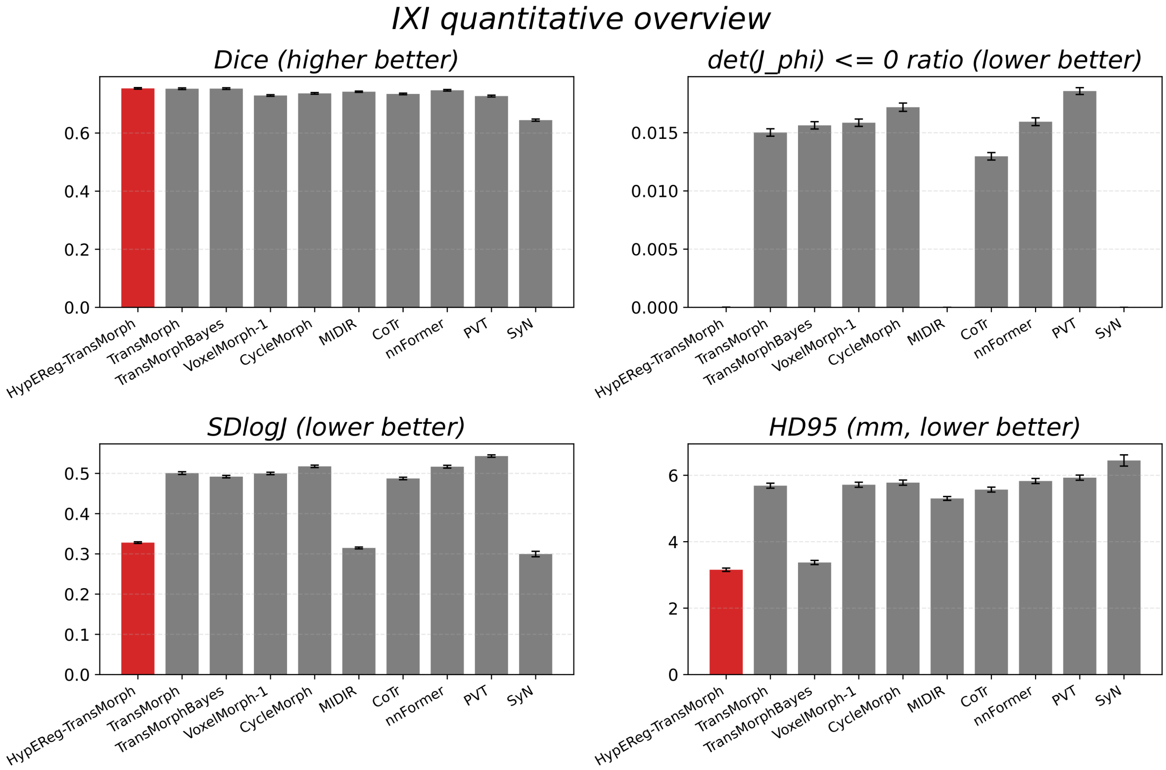

Figure S1: Descriptive IXI quantitative overview (Dice, non-positive Jacobian determinant ratio, SDlogJ, and HD95). Bars include standard-error error bars for visual uncertainty context. (HypEReg-TransMorph is abbreviated as HER-TransMorph in the bar-chart panel labels; **HER** $\equiv$ **HypEReg** throughout.)

3 Figure S2: Qualitative Registration Comparison

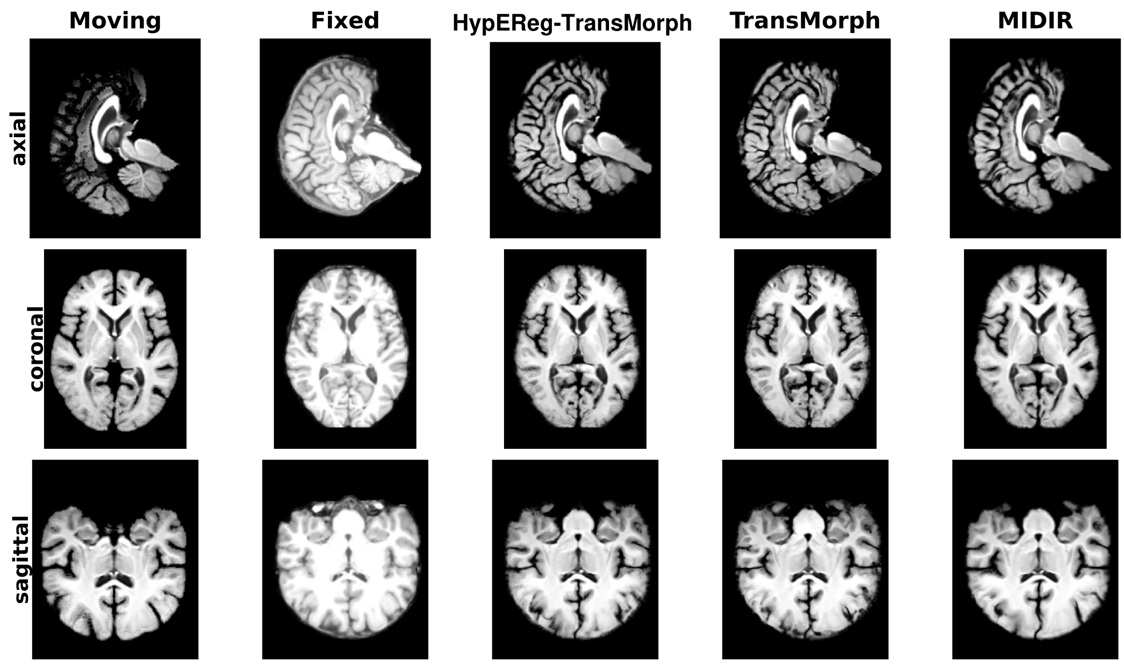

Figure S2: Qualitative registration comparison on a representative IXI test subject across three orthogonal views (axial/coronal/sagittal). Columns show moving atlas, fixed target, and warped outputs from HypEReg-TransMorph, TransMorph, and MIDIR. Boundary-alignment behaviour in ventricles, cortex, and deep gray structures complements the deformation-grid and Jacobian evidence presented in the main text. Column titles in the figure panels use the abbreviated name HypEReg-TransMorph for the proposed method.

4 Inferential Statistics (HypEReg-TransMorph vs. Baselines)

Metric definitions (to avoid naming ambiguity). In this Supplementary section, `dice_mean` denotes the per-subject mean Dice over the full 46-label FreeSurfer-compatible label set in the per-case metric tables, whereas the main paper reports *grouped Dice* averaged over 17 grouped VOIs (Table 2 in the main text). The two numbers therefore have different scales and are not directly comparable.

Table S2: Auxiliary deformation and surface metrics on IXI (mean±std over subjects). This table reports NSD@1 mm and additional deformation-regularity descriptors that are referenced in the main text but not included in Table 2.

Model	**NSD@1 mm**↑	**Bending**↓	**MADiv**↓	$J_{\min}$	J_{p01}	J_{p99}	$J_{\max}$
HypEReg-TransMorph	0.8480 ± 0.0317	0.0031 ± 0.0001	0.1772 ± 0.0114	-0.2700 ± 0.1496	0.2754 ± 0.0268	2.1284 ± 0.0669	18.6155 ± 2.9607
TransMorphBayes	0.8462 ± 0.0347	0.0050 ± 0.0003	0.2341 ± 0.0114	-6.6766 ± 2.2896	-0.0871 ± 0.0555	2.5926 ± 0.0764	46.0976 ± 9.2892
HypEReg (volume only)	0.7715 ± 0.0398	0.0036 ± 0.0001	0.1719 ± 0.0077	-0.1754 ± 0.0993	0.3433 ± 0.0196	2.1054 ± 0.0979	12.3397 ± 2.2249
HypEReg+Grad (volume only)	0.7612 ± 0.0403	0.0019 ± 0.0001	0.1549 ± 0.0065	-0.1014 ± 0.0928	0.3729 ± 0.0187	2.0257 ± 0.0840	8.4640 ± 1.1222
HypEReg+Grad (fold only)	0.7242 ± 0.0424	0.0041 ± 0.0001	0.2367 ± 0.0064	-1.2721 ± 0.3317	0.0776 ± 0.0232	2.5189 ± 0.0782	19.8379 ± 2.8103

Bending = voxel-averaged bending energy; MADiv = mean absolute divergence of the displacement field; NSD@1 mm computed with $\tau = 1$ mm. "HypEReg (volume only)" disables the fold hinge ($\gamma = 0$); "HypEReg+Grad" rows include the standard first-order smoothness term $\mathcal{L}_{\text{grad}}$ and then isolate either the volume component ($\gamma = 0$) or the fold component ($\beta = 0$) of HypEReg. Values are consolidated evaluation exports for the listed models.

Table S3: Inferential summary for TransMorphBayes and MIDIR (mean±std, aligned median difference, Wilcoxon p, Benjamini–Hochberg q, and matched-pairs rank-biserial effect). HypEReg-TransMorph denotes the proposed model. Where `dice_mean` appears, values are *not* the main-text grouped Dice over 17 VOIs (Table 2); see table footnote.

Metric	**Baseline**	**HypEReg mean±std**	**Baseline mean±std**	**Median diff (aligned)**	p/q**/effect**
dice_mean	TransMorphBayes (uploaded ckpt)	0.5765±0.0301	0.5731±0.0313	0.00280	6.58×10^{-10} / 6.58×10^{-10} / 0.6636
non_jec	TransMorphBayes (uploaded ckpt)	$1.46 \times 10^{-5} \pm 7.37 \times 10^{-6}$	0.01510±0.00329	0.01489	1.31×10^{-20} / 3.28×10^{-20} / 1.0000
SDlogJ	TransMorphBayes (uploaded ckpt)	0.3280±0.0221	0.4934±0.0334	0.16325	1.31×10^{-20} / 3.28×10^{-20} / 1.0000
HD95_mean	TransMorphBayes (uploaded ckpt)	5.3234±0.6936	5.7246±0.7604	0.33877	1.77×10^{-18} / 2.94×10^{-18} / 0.9424
ASSD_mean	TransMorphBayes (uploaded ckpt)	1.3570±0.1770	1.4160±0.1832	0.06143	2.98×10^{-11} / 3.72×10^{-11} / 0.7142
dice_mean	MIDIR	0.5765±0.0301	0.5643±0.0244	0.01166	3.79×10^{-18} / 7.76×10^{-18} / 0.9331
non_jec	MIDIR	$1.46 \times 10^{-5} \pm 7.37 \times 10^{-6}$	0.0000±0.0000	-1.31×10^{-5}	1.31×10^{-20} / 3.69×10^{-20} / -1.0000
SDlogJ	MIDIR	0.3280±0.0221	0.3148±0.0242	-0.01365	4.84×10^{-20} / 1.28×10^{-19} / -0.9850
HD95_mean	MIDIR	5.3234±0.6936	5.3028±0.6139	-0.00855	0.5767 / 0.5898 / -0.0600
ASSD_mean	MIDIR	1.3570±0.1770	1.4100±0.1585	0.06514	3.58×10^{-9} / 4.36×10^{-9} / 0.6342

Aligned median difference is oriented so positive values indicate HypEReg-TransMorph improvement for the given metric direction. The effect is matched-pairs rank-biserial correlation. Pair counts: $n = 115$ for both baselines in this table. **`dice_mean`:** per-subject mean Dice averaged over the full 46-label evaluation set (same convention as the metric-definition paragraph above). Main-text overlap uses *grouped* Dice (mean over 17 aggregated VOIs). The two scalings differ, so `dice_mean` means/std/median contrasts must not be compared numerically to grouped-Dice entries in Table 2; Wilcoxon p and BH q on `dice_mean` rows test paired differences on this 46-label convention only.

Table S4: Selected paired comparisons (Wilcoxon signed-rank with Benjamini–Hochberg FDR). Positive signed effect indicates HypEReg-TransMorph improvement under metric-aligned directionality; signed effect is matched-pairs rank-biserial correlation. Rows with `dice_mean` follow the 46-label per-subject mean convention (not main-text grouped Dice); see table footnote.

Metric	Baseline	Median paired diff (aligned)	p-value	q-value	Signed effect
non_jec	TransMorph	0.014747	1.31×10^{-20}	3.69×10^{-20}	1.0000
SDlogJ	TransMorph	0.170655	1.31×10^{-20}	3.69×10^{-20}	1.0000
HD95_mean	TransMorph	0.319583	3.77×10^{-17}	7.08×10^{-17}	0.9046
ASSD_mean	TransMorph	0.051747	6.09×10^{-11}	7.83×10^{-11}	0.7028
non_jec	MIDIR	-0.000013	1.31×10^{-20}	3.69×10^{-20}	-1.0000
SDlogJ	MIDIR	-0.013652	4.84×10^{-20}	1.28×10^{-19}	-0.9850
HD95_mean	MIDIR	-0.008554	0.5767	0.5898	-0.0600
ASSD_mean	MIDIR	0.065138	3.58×10^{-9}	4.36×10^{-9}	0.6342
non_jec	TransMorph (uploaded ckpt)	0.014747	1.31×10^{-20}	2.63×10^{-20}	1.0000
dice_mean	TransMorph (uploaded ckpt)	0.002326	6.88×10^{-7}	6.88×10^{-7}	0.5334
non_jec	TransMorphBayes (uploaded ckpt)	0.014714	1.31×10^{-20}	2.63×10^{-20}	1.0000
dice_mean	TransMorphBayes (uploaded ckpt)	0.002706	1.74×10^{-10}	2.32×10^{-10}	0.6858
non_jec	SyN	-0.000012	8.00×10^{-16}	1.24×10^{-15}	-0.8654
SDlogJ	SyN	-0.045599	3.09×10^{-5}	3.31×10^{-5}	-0.4477
HD95_mean	SyN	0.705730	1.31×10^{-7}	1.47×10^{-7}	0.5670
ASSD_mean	SyN	-0.111801	0.6572	0.6572	0.0477

`dice_mean` rows: paired tests use per-subject mean Dice over 46 labels (export field name), which is on a different absolute scale than grouped Dice in main-text Table 2; do not equate magnitudes across documents. All other metrics match the main evaluation pipeline. Pair count $n = 115$ where full paired exports exist.

Full inferential outputs across all exported metric-baseline pairs are provided in the supplementary release package. Additional uploaded-checkpoint paired results for TransMorph and TransMorphBayes refer to `dice_mean` (46-label mean per subject, not grouped Dice) and non-positive Jacobian ratio, included with corresponding subject-level inputs for completeness. Repeated values such as $p = 1.31 \times 10^{-20}$ occur when the two-sided Wilcoxon test reaches the practical floating-point floor under near-complete sign consistency for $n = 115$, and should be interpreted as extremely small p-values.

5 Table S5: Grouped-VOI Mapping

Table S5: Grouped-VOI mapping used for grouped Dice reporting.

Grouped VOI	FreeSurfer-compatible label IDs
Brain-Stem	16
Thalamus	10, 49
Cerebellum-Cortex	8, 47
Cerebral-White-Matter	2, 41
Cerebellum-White-Matter	7, 46
Putamen	12, 51
VentralDC	28, 60
Pallidum	13, 52
Caudate	11, 50
Lateral-Ventricle	4, 43
Hippocampus	17, 53
3rd-Ventricle	14
4th-Ventricle	15
Amygdala	18, 54
Cerebral-Cortex	3, 42
CSF	24
choroid-plexus	31, 63

6 OASIS Cross-Cohort Training and Environment

Primary OASIS structural MRI are distributed through the public portal `https://www.oasis-brains.org/` (Open Access Series of Imaging Studies). Train/validation/test splits and preprocessing match the Learn2Reg challenge cross-subject registration setup for OASIS; see the main manuscript Methods for citations and for the definition of the preprocessed release used in our experiments.

OASIS in-cohort retraining mirrors the IXI HypEReg-TransMorph recipe: the same TransMorph backbone as above, Adam with AMSGrad (10^{-4} learning rate, weight decay 0), polynomial LR decay with exponent 0.9 over 500 epochs, bidirectional updates, batch size 2 on the OASIS train split, and HypEReg weights $(\beta, \gamma, \epsilon) = (0.02, 20, 10^{-3})$. Use Python/PyTorch builds compatible with the host GPU and the same antspyx/SimpleITK stack as in the IXI evaluation where classical baselines are compared. The released artifact includes the OASIS-retrained reference model used for Supplementary Table S6; it is *not* the strict IXI→OASIS zero-shot checkpoint reported in the main text.

7 Table S6: OASIS Paired Wilcoxon / BH-FDR

Reference model: OASIS-retrained HypEReg-TransMorph. All pairs n=19. Only significant contrasts ($q_{\mathrm{BH}} < 0.05$) shown; sign of direction: + = HypEReg-TransMorph better. Unless otherwise stated, baselines in this table are also *OASIS in-cohort retrained* models evaluated on the same 19 OASIS test pairs. This table is therefore distinct from the main-text IXI→OASIS zero-shot transfer results.

Table S6: Significant paired Wilcoxon contrasts (BH-FDR, $q < 0.05$) for HypEReg-TransMorph vs baselines on 19 OASIS test pairs.

Baseline	**Metric**	**HypEReg mean**	**Baseline mean**	p / q
CycleMorph	dice_mean↑	0.7966	0.7243	3.8×10^{-6} / 3.8×10^{-6}
	non_jec↓	1.0×10^{-5}	8.2×10^{-3}	3.8×10^{-6} / 3.8×10^{-6}
	SDlogJ↓	0.2786	0.4345	3.8×10^{-6} / 3.8×10^{-6}
	HD95_mean↓	2.294	3.076	3.8×10^{-6} / 3.8×10^{-6}
	ASSD_mean↓	0.680	0.949	3.8×10^{-6} / 3.8×10^{-6}
MIDIR	dice_mean↑	0.7966	0.7254	3.8×10^{-6} / 3.8×10^{-6}
	non_jec↓	1.0×10^{-5}	0.0000	1.3×10^{-4} / 1.3×10^{-4} (*HypEReg worse*)
	SDlogJ↓	0.2786	0.2551	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
	HD95_mean↓	2.294	2.893	3.8×10^{-6} / 3.8×10^{-6}
	ASSD_mean↓	0.680	0.931	3.8×10^{-6} / 3.8×10^{-6}
TransMorph (dsc0.857)	dice_mean↑	0.7966	0.8610	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
	non_jec↓	1.0×10^{-5}	8.1×10^{-3}	3.8×10^{-6} / 3.8×10^{-6}
	SDlogJ↓	0.2786	0.4211	3.8×10^{-6} / 3.8×10^{-6}
	HD95_mean↓	2.294	1.598	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
	ASSD_mean↓	0.680	0.483	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
VoxelMorph-1	dice_mean↑	0.7966	0.7159	3.8×10^{-6} / 3.8×10^{-6}
	non_jec↓	1.0×10^{-5}	8.1×10^{-3}	3.8×10^{-6} / 3.8×10^{-6}
	SDlogJ↓	0.2786	0.4291	3.8×10^{-6} / 3.8×10^{-6}
	HD95_mean↓	2.294	3.181	3.8×10^{-6} / 3.8×10^{-6}
	ASSD_mean↓	0.680	0.963	3.8×10^{-6} / 3.8×10^{-6}

p: two-sided Wilcoxon signed-rank; q: Benjamini–Hochberg FDR. "HypEReg worse" indicates the signed direction favours the baseline on that metric; these contrasts remain significant, indicating MIDIR/dsc0.857 significantly outperform HypEReg-TransMorph on those regularity/overlap dimensions. Reported non_jec values at or near 10^{-5} reflect the evaluation export precision floor on this OASIS split (with MIDIR exactly 0 by construction under B-spline parameterization), so Wilcoxon significance is driven by paired subject-level ranks rather than by visual separation of rounded means alone. Full numeric output accompanies the software release.

8 Table S7: Bootstrap 95% Confidence Intervals (IXI)

Table S7: Bootstrap 95% confidence intervals for IXI subject-level metric means ($B = 10{,}000$, sampling with replacement, seed = 0, $n = 115$ per model).

Model	Dice mean [95% CI]	$\det(J_\phi) \leq 0$ ratio mean [95% CI]	SDlogJ mean [95% CI]	HD95 mean [95% CI]	ASSD mean [95% CI]
HypEReg-TransMorph	0.7537 [0.7486, 0.7587]	1.461×10^{-5} [1.332×10^{-5}, 1.604×10^{-5}]	0.3280 [0.3240, 0.3322]	5.3234 [5.1952, 5.4511]	1.3570 [1.3247, 1.3897]
TransMorph	0.7527 [0.7470, 0.7581]	1.502×10^{-2} [1.441×10^{-2}, 1.566×10^{-2}]	0.5064 [0.4943, 0.5076]	5.6872 [5.5391, 5.8376]	1.4073 [1.3735, 1.4429]
TransMorphBayes	0.7530 [0.7474, 0.7584]	1.510×10^{-2} [1.451×10^{-2}, 1.571×10^{-2}]	0.4920 [0.4874, 0.4996]	5.7246 [5.5847, 5.8646]	1.4160 [1.3831, 1.4501]
MIDIR	0.7423 [0.7380, 0.7465]	0 [0, 0]	0.3148 [0.3105, 0.3194]	5.3028 [5.1898, 5.4172]	1.4100 [1.3815, 1.4395]
SyN (ANTs)	0.6445 [0.6372, 0.6518]	5.874×10^{-6} [1.235×10^{-6}, 1.243×10^{-5}]	0.2996 [0.2866, 0.3130]	6.4457 [6.1174, 6.7784]	1.5233 [1.3838, 1.6675]

Dice follows the grouped 17-structure protocol used in the main paper; non-positive Jacobian ratio, SDlogJ, HD95, and ASSD come from the same per-case evaluation pipeline. Bootstrap: $B = 10{,}000$, sampling with replacement, seed 0.